TRANSDUCERS

Instrumentation system: An instrumentation system consists of a number of components used to perform a measurement and record the result. It consists of three major elements: an input devices, a signal conditioning or processing device, and an output device. The input device receives the quantity under measurement and delivers a proportional electrical signal to the signal-conditioning device. Here the signal is amplified, filtered, or modified to a format acceptable to the output device. The output device may be a simple indicating meter, an oscilloscope, or a chart recorder for visual display. It may be a magnetic tape recorder for temporary or permanent storage of the input data, or it may be a digital computer for data manipulation or process control.

Transducer - Is a device that converts non-electrical quantities into an electrical signal. Such non-electrical quantities are heat, light intensity, humidity. Others such as mechanical force or displacement.

Transducers may be classified according to their application, method of energy conversion, nature of the output signal etc.
Transducers are classified as passive transducers and self generating transducers.

ELECTRICAL TRANSDUCERS

Electrical transducers converts non-electrical quantities into electrical quantities first and then measured.

TYPES OF ELECTRICAL TRANSDUCERS: Various types of Electrical transducers as follows:

(i) Resistive type (ii) Capacitive type (iii) Inductive type.

1. Resistive type Transducer - In such a transducer, resistance between the output terminals of a transducer gets varied according to the measurand.

Resistance of any metal conductor is given by the equation

$$R = \rho \frac{l}{a} \quad \ldots (1)$$

Where ρ is the resistivity of the material of conductor in Ω/m,
l is the length of conductor in meters and
a is the cross-sectional area of conductor in m^2.

Physical phenomenon i.e input signal to the transducer causes variation in resistance by changing any one of the quantities ρ, l and a. For measurement of displacement length of conductor is varied in potentiometer thereby resulting in change in resistance. For measurement of force and pressure resistance of a conductor or semiconductor is varied in strain gauge. Variation in temperature causes change in resistivity of the conductor. With some devices resistance varies with the change in light intensity because of photo conductive effect, while with others resistance varies on exposure to magnetic field due to magnetoresistance effect.

2. Inductive transducers - These transducers operate generally upon one of the following three principles

(i) Variation of self inductance of the coil
(ii) Variation of Mutual Inductance of the coil.
(iii) Provision of eddy Current.

(i) Transducer operating on the principle of variation of self-inductance

The self inductance of a coil is given by the equation

$$L = \frac{N^2}{l/\mu A} = N^2 \frac{\mu A}{l} = N^2 \mu G \quad \ldots (2)$$

Where N is the number of turns on the coil,
l ~~is~~ the mean length of the magnetic path.
A is the area of the ~~cross-sectional~~ of the magnetic path.
μ is the permeability of the magnetic material.
A/l is called the geometric form factor; G.

From the above equation it shows that the self inductance of a coil can be varied by varying the number of turns on the coil, the permeability of the magnetic material or by changing the geometrical configuration of the magnetic circuit.
These transducers are usually used for measurement of displacement both linear and angular displacement.

(ii) Transducers operating on the principle of Variation of Mutual Inductance.

Such transducers operate on the fact that mutual inductance between two coils depends upon the self inductance of the coils and coefficient of coupling between them, as mutual inductance between two coils is given by the equation

$$M = K\sqrt{L_1 L_2} \quad \ldots (3)$$

Where K is the coefficient of coupling and L_1 and L_2 are self inductances of the coils.
In such transducers two or more coils are used and the mutual inductance between the coils is varied as ~~measu~~ measurand.

(iii) Transducers operating on the principle of production of eddy currents.
These type of transducers operate on the principle that when a conducting plate is kept near a coil carrying

alternating current, eddy currents are induced in the conducting plate, producing its own magnetic field in opposition to the main field created by the coil. Thus eddy current induced in the conducting plate reduce the net flux linking with the coil and so the inductance of the coil is reduced. The nearer is the plate to the coil, the higher are the ~~eddy~~ induced eddy currents and so higher is the reduction in the inductance of the coil. Thus, the inductance of the coil changes with the movement of the plates.

3. Capacitive Transducers

The variable capacitance transducer comprises of a capacitor, the capacitance of which is varied by the non-electrical quantity being measured.

The capacitance of a parallel plate capacitor, C is given by an equation

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d} \text{ Farads} \qquad \text{---(4)}$$

Where ε_0 is the permittivity of free space and is equal to 8.854×10^{-12} F/M;

ε_r is the relative permittivity of the dielectric,

A is the area of plates in M^2 and

d is the distance between the two plates in mets.

As the capacitance of a capacitor depends upon the area of its electrodes, the distance between them and the relative permittivity of the dielectric material, hence such transducer can be used for measurement of non-electrical quantities by affecting any one of the above mentioned parameters.